Data Structures & Analysis of Algorithms

Dr. Rabins Porwal

Suggested Readings

- *Text Books:*
  - E. Horowitz and S. Sahani, "*Fundamentals of Data Structures in C*", 2nd Edition, Universities Press, 2008.
  - Mark Allen Weiss, "*Data Structures and Algorithm Analysis in C*", 2nd Edition Addison-Wesley, 1997.
  - Mary E. S. Loomis, "*Data Management and File Structure*", PHI, 2nd Ed., 2009.
- *References:*
  - Schaum's Outline Series, "*Data Structure*", TMH, Special Indian Ed., Seventeenth Reprint, 2009.
  - Y. Langsam et. al., "*Data Structures using C and C++*", PHI, 1999.
  - N. Dale and S.C. Lilly, D.C. Heath and Co., "*Data Structures*", 1995.
  - R. S. Salaria, Khanna, "*Data Structure & Algorithms*", Book Publishing Co. (P) Ltd., 2002.
  - Richard F. Gilberg and Behrouz A. Forouzan, "*Data Structure: A Pseudocode Approach with C*", Cengage Learning, 2nd Ed., 2005.

Objectives

- To identify and create useful mathematical entities and operations to determine what classes of problems can be solved by using these entities and operations.
- To determine the *representation* of these abstract entities and to *implement* the abstract operations on these concrete representations.

Definition

- Data structure is the branch of computer science that unleashes the knowledge of *how the data should be organized*, *how the flow of data should be controlled* and how this structure (representation) should be designed and implemented *to reduce the complexity* and *increase efficiency of the algorithm.*

Need of a Data Structure

- It helps in understanding the relationship of one data element with the other and organize it within the memory.
  - Simple – linear relationship
    - Ex: *list of names of moths in a year (overruled in case of first and last month's names)*
  - Complex – hierarchical or graphical relationship
    - Ex: *location of historical places in a country*
- Thus data structure helps in analyzing the data, storing it and organizing it in a logical or mathematical manner.

Abstract Data Types (ADT)

- A mathematical model of the data objects that make up a data type as well as the functions that operate on these objects.
- It is the specification of logical and mathematical properties of a data type or structure.
- It acts as a useful guideline to implement a data type correctly.

Data Types

- A method of interpreting a bit pattern is often called a data type.
- It refers to the implementation of the mathematical model specified by an ADT
- It is the concept defined by a set of logical properties.
- In general, a data type can be defined as a concept that defines an internal representation of data in the memory.

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Primitive Data Types

- These are basic data types of any language that form the basic unit for the data structure defined by the user.
- It defines how the data will be internally represented in, stored, and retrieved from the memory.
  - Integer
  - Char
  - Real/ Float Numbers

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Difference bet. ADT, Data Types and Data Structures

- An ADT is the specification of the data type which specifies the logical and mathematical model of the data type.
- A data type is the implementation of an abstract data type.
- Data structure refers to the collection of computer variables that are connected in some specific manner.

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Data Organizations/ Structures

- Arrays
- Linked Lists
- Trees
- Another types of data structures
  - Stack
  - Queue
  - Graph

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Data Structure Operations

- Major Four operations
  - Traversing – accessing each record exactly once so that certain items in the record may be processed/ visited
  - Searching – finding the location of the record with a given key value, or finding the locations of all records which satisfy one or more conditions.
  - Inserting – adding a new record to the structure.
  - Deleting – removing a record from the structure.

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Data Structure Operations (Contd.)

- Two more operations, which are used in special situations.
  - Sorting – arranging the records in some logical order (ascending/ descending using numerical value, or alphabetical arrangement)
  - Merging – combining the records of two different sorted files into a single sorted file

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